# RIPHAH INTERNATIONAL UNIVERSITY




## Faculty of Computing
## FINAL YEAR PROJECT PROPOSAL

# SensoryGuard

### Project Team


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# Project Proposal

**Project Title:** SensoryGuard: Real-Time Multimodal Background Safeguard for Short-Form Videos

## 1. Introduction:

The rapid growth of short-form video, particularly YouTube Shorts, has changed how children under 12 in Pakistan consume media. Although the platform contains educational content, its recommendation system can also expose children to highly stimulating videos, rapid edits, violent visuals and abusive language in Urdu, Roman Urdu, and English. Such localized content may bypass standard moderation filters, exposing children to profanity, aggressive slang, and vulgarity during important stages of cognitive and speech development.

**SensoryGuard** is an automated Android application designed to protect children from harmful sensory and linguistic content in real time. Running alongside YouTube, it uses multimodal analysis to monitor rapid scene changes, sudden audio spikes and speech in Urdu, Roman Urdu, and English. The system calculates a real-time **Sensory Harm Index (SHI)** and can respond by blurring the screen or muting audio when harmful content is detected.

## 2. Problem Statement:

Children under 12 in Pakistan are often exposed to harmful content on YouTube Shorts because existing safety filters do not effectively detect localized risks. Profanity, slang, and abusive language in Urdu, Roman Urdu and English can easily go unnoticed. Combined with rapid video cuts, sudden loud sounds, and psychological hooks such as **end tak dekho**, this content can contribute to mental fatigue, reduced attention spans and behavioral problems among young viewers. Therefore, there is a need for an AI driven multimodal solution that can process visual, auditory and linguistic cues simultaneously to tackle this problem in real time.

## 3. Literature Review:

### 3.1. Market Survey:

**Table 1: Summary of Market Survey**

| Feature | YouTube / YouTube Kids | Google Family Link | AirDroid Parental Control | Kids360 | Qustodio | SensoryGuard (Proposed) |
|---|---|---|---|---|---|---|
| **Real-time analysis of currently playing Short** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Visual violence detection** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Audio volatility tracking** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Speech-to-text analysis** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Urdu / Roman Urdu offensive-language detection** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Individual-Short intervention** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Black overlay with swipe-up warning** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| **Parent report of detected harmful events** | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |

## 3.2. Related Research Work

### 3.2.1. Research Work 1

**(Papadamou, et al., 2020)** studied inappropriate YouTube videos targeting young children and developed methods to characterize and detect such content. The study showed that YouTube's recommendation system can expose children to unsuitable material. However, the work focused on detecting inappropriate videos at the platform level rather than providing a local Android system that can intervene in real time while a Short is playing.

### 3.2.2. Research Work 2

**(Jo & Wojcieszak, 2025)** introduced MetaHarm, a multimodal YouTube harm dataset containing 19,422 videos annotated by domain experts, GPT-4-Turbo, and crowdworkers. The dataset uses harmful/harmless labels across six categories, including hate and harassment, addictive, sexual, and physical harms. The work supports the use of multimodal data for detecting harmful YouTube content. However, MetaHarm is a dataset and measurement resource rather than a real-time Android system that can intervene during video playback.

### 3.2.3. Research Work 3

**(Xue, et al., 2026)** analyzed 4,492 short videos recommended to children and teenagers on YouTube Shorts, TikTok, and Instagram Reels. The study examined audiovisual and thematic features, including shot changes and frame-level visual characteristics, and identified differences between safe and unsafe videos. This work is closely related to SensoryGuard, but it focuses on analyzing short-video content rather than developing an Android overlay that can intervene on the Short currently playing.

### 3.2.4. Research Work 4

**(Thanh, Hung, Nguyen, & Trong, 2025)** proposed MTikGuard, a real-time multimodal system for detecting harmful content on TikTok. It combines video, audio, and text analysis and reported 89.37% accuracy and 89.45% F1 score in its test setting. The system uses Kafka and Spark for large-scale streaming. However, MTikGuard is designed for TikTok and server-side streaming, while SensoryGuard targets YouTube Shorts on Android and focuses on real-time intervention on the individual Short being viewed.

### 3.2.5. Research Work 5

**(Rizwan, Shakeel, & Karim, 2020)** introduced RUHSOLD, a dataset of 10,012 Roman Urdu tweets with coarse- and fine-grained labels for hate speech and offensive language. The study showed that transfer learning and domain-specific resources can help detect harmful content in Roman Urdu. However, the dataset consists of social media text, so it would need to be adapted before being used to detect harmful speech in YouTube Shorts.

### 3.2.6. Research Work 6

**(Akhter, Jiangbin, Naqvi, Abdelmajeed, & Sadiq, 2020)** developed offensive-language datasets for Urdu and Roman Urdu, including 2,171 Urdu comments and a 10,000-comment Roman Urdu subset collected from YouTube and other social media sources. Using character and word n-grams, the study achieved strong results in offensive-language classification. The datasets provide useful resources for local-language detection, but the work does not combine language analysis with video and audio features or support real-time intervention.

**Table 2: Summary of Related Research Work**

| Work | Main input | Main result / contribution | Limitation for SensoryGuard |
|---|---|---|---|
| **Papadamou et al.** | YouTube videos | Child-focused inappropriate-content detection | No Android individual-short intervention |
| **MetaHarm** | Video + text + frames | Large multimodal YouTube harm dataset | Dataset, not a device-side filter |
| **Xue et al.** | Short videos | Audiovisual markers of unsafe Shorts | Analysis, not live intervention |
| **MTikGuard** | Video + audio + text | Real-time TikTok moderation | Different platforms and deployment |
| **Rizwan et al.** | Roman Urdu text | Hate/offensive language labels | Text only |
| **Akhter et al.** | Urdu/Roman Urdu text | Offensive-language datasets | Text only |

## 3.3. Research Gap

The market survey shows that existing parental-control tools mainly manage access, screen time, apps, websites, and viewing history. Some can report YouTube activity, but the reviewed tools do not provide the full functionality required for this project: analyzing the current Short's video frames, audio, and speech, determining whether it is harmful, and intervening on that specific Short.

Research has moved further toward content-level analysis. YouTube studies have examined inappropriate videos and harmful-content datasets, while recent short-video research has analyzed visual, audio, and thematic features linked to unsafe content. TikGuard and MTikGuard also show that transformer-based video classification and multimodal analysis can support short-video safety. However, these systems either target

other platforms or focus on classification and large-scale processing rather than a lightweight Android safety layer for YouTube Shorts.

Another gap is local-language support. Urdu and Roman Urdu offensive-language datasets are available, but most are text-based rather than speech datasets collected from YouTube Shorts. This creates a need to convert speech to text and classify it using models trained or adapted for Urdu and Roman Urdu.

Based on these findings, SensoryGuard addresses four gaps in one system: YouTube Shorts as the target platform, real-time analysis of the current Short, Urdu and Roman Urdu language support, and Android-based intervention that affects only the detected Short instead of blocking YouTube entirely.

## 3.4. Proposed Novelty

SensoryGuard proposes an Android safety layer for YouTube Shorts. When a Short starts, the application captures the required screen and playback data with user permission, divides the input into short windows, and runs video, audio, and language analysis. The results are combined into a risk decision. If the risk exceeds a selected threshold, SensoryGuard displays a warning overlay and covers the current Short. Where Android and the source app allow playback capture, the system can also mute the captured audio during the warning. Android's MediaProjection API requires explicit user permission and a media-projection foreground service, while audio playback capture depends on the source app's capture policy.

SensoryGuard also maintains a session report containing the event time, detected risk type, confidence score, and action taken. This allows parents to review flagged events without watching the child's entire session.

The system does not assume that any single video feature proves psychological harm. Visual pacing and audio changes are treated as measurable signals that contribute to the overall content-risk decision, while defined categories such as violence and offensive language are treated as content classes.

# 4. Scope

The proposed system is limited to YouTube Shorts on Android smartphones and tablets. The first prototype will focus on content that can be assessed from the Short itself and will not attempt to control the entire Android device or all websites.

- Detect potentially harmful visual content, with violence as the main visual category.
- Measure visual changes between sampled frames to identify unusually rapid scene changes or pacing.
- Measure audio level variation and sudden peaks as supporting sensory signals.

- Convert speech from the Short into text and check English, Urdu and Roman Urdu for offensive or toxic language.
- Combine the outputs of the visual, audio and language checks into a risk decision.
- Show an overlay warning and cover the current Short when the decision crosses the selected threshold.
- Allow a safe next Short to continue without blocking the YouTube application.
- Mute playback during an intervention when Android capture and device conditions permit it.
- Store a parent-facing session report containing detected category, time and action.
- Measure accuracy, precision, recall, F1-score, false positives, detection delay and end-to-end intervention latency during testing.

## 5. Datasets

The datasets will not be treated as one large mixed dataset. Each source will first be mapped to the part of the problem it can solve. The final training and validation sets will use common labels, and a separate test set will be kept for final evaluation.

| Dataset / source | Type | Use in SensoryGuard | Planned use |
|---|---|---|---|
| **Real Life Violence Situations** | Video | Violence / non-violence | Train and test visual violence model |
| **TikHarm** | Short video | Safe, harmful, adult, suicide | Supplement child-focused video safety data |
| **TIKHARM Kaggle** | Short video | Same TikHarm source | Use only if licensing/source is confirmed |
| **Roman-Urdu-Toxic-Corpus** | Text | Toxic / non-toxic Roman Urdu | Train language classifier |
| **RUHSOLD** | Text | Hate / offensive / normal | Add Roman Urdu language labels |
| **Urdu Abusive Dataset** | Text | Urdu offensive / non-offensive | Support Urdu script classification |
| **ROU-AudioTox** | Audio + text | Roman Urdu offensive speech | Candidate speech evaluation/training source after access check |

The Real-Life Violence Situations dataset contains 1,000 violence and 1,000 non-violence videos collected from YouTube, making it suitable for training the initial visual violence classifier. TikHarm contains 3,948 short videos across four categories and was developed for child-safety research on TikTok. It can support harmful-content classification but should not be presented as YouTube data. The Roman Urdu Toxic Corpus contains 72,771 labeled samples and can be used for the language-analysis branch.

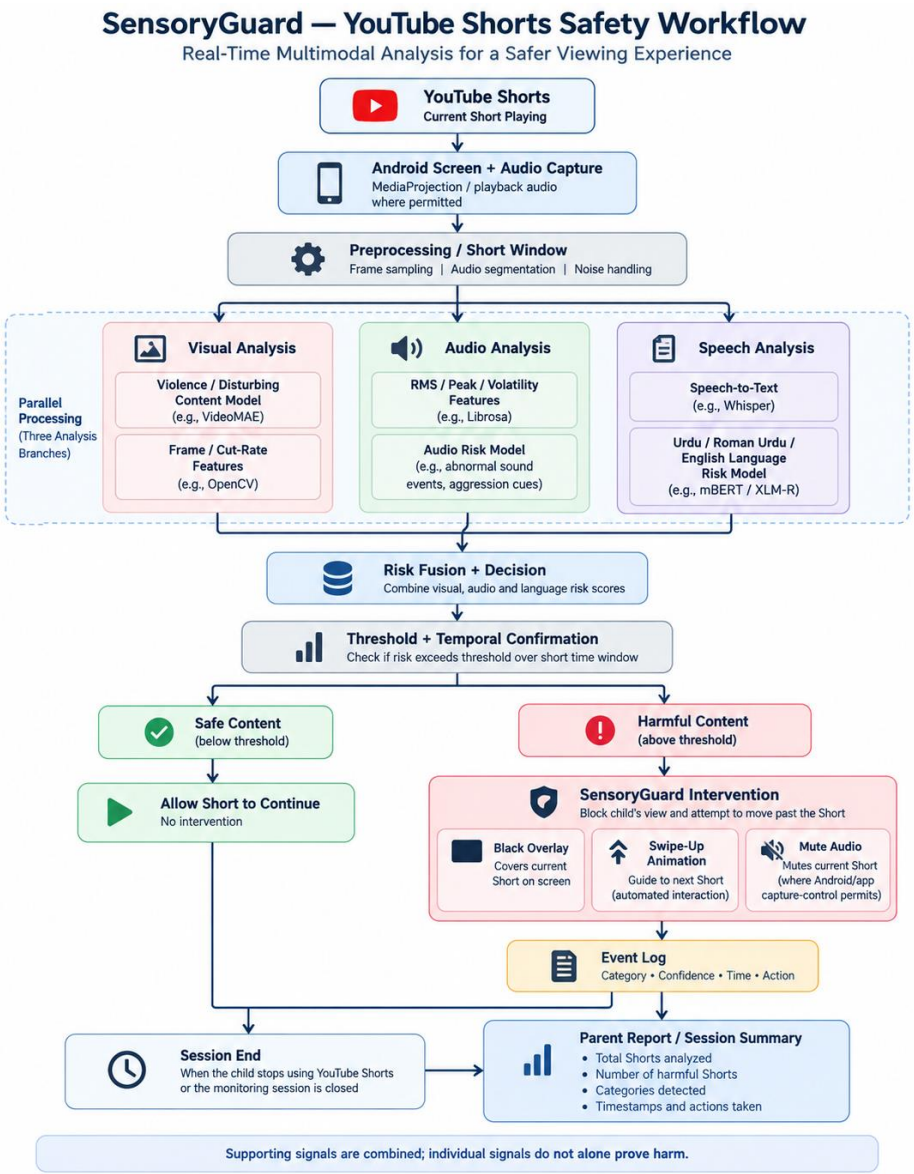
## 6. Workflow

The system will work as a continuous process. The aim is to make a decision from a short window of the currently playing Short instead of waiting for the complete video to finish.

- **YouTube Short starts:** The child opens YouTube Shorts or scrolls to a new Short.
- **Capture:** SensoryGuard uses Android MediaProjection to capture the currently displayed screen and, where permitted, playback audio for analysis. The user must grant screen-capture permission, and current Android versions impose additional requirements for MediaProjection foreground services. Audio capture also depends on the source app allowing playback capture.
- **Prepare input:** The captured video is divided into short processing windows. Frames are sampled for visual analysis, audio is divided into short windows for audio analysis, and available speech is passed to the speech-to-text component.
- **Parallel analysis:** The three analysis branches run in parallel to reduce unnecessary processing delay:
    - **Visual analysis:** A video model estimates violence or disturbing visual content, while OpenCV measures frame changes and scene-cut rate.
    - **Audio analysis:** RMS and peak-level changes are measured as supporting audio signals for unusual or aggressive sound patterns.
    - **Language analysis:** Speech is converted to text, normalized, and classified for offensive or toxic language in English, Urdu, and Roman Urdu.
- **Combine results:** The visual, audio, and language results are combined using the selected decision rule. Temporal confirmation is applied so that one noisy frame, sound, or word does not immediately trigger an intervention.
- **Decision:** If the combined risk remains above the selected threshold, the Short is treated as potentially harmful. If the risk remains below the threshold, the Short continues normally.
- **Intervention:** For a potentially harmful Short, SensoryGuard places a black overlay over the current Short, preventing the child from continuing to view it, and displays a swipe-up animation instructing the child to move to the next Short. SensoryGuard can use Android's AccessibilityService capabilities to display an accessibility overlay and, where permitted and properly configured, perform a swipe gesture.
- **Audio intervention:** SensoryGuard attempts to mute the Short's audio where the Android device and source application allow the required audio-control behavior. This

should be described as an attempted audio intervention, rather than claiming that MediaProjection itself can mute YouTube. Android's playback-capture facility provides access to audio for capture under its applicable rules; it does not by itself give an app unrestricted control over another app's playback.

- **Continue or log:** Safe Shorts continue normally. When a potentially harmful Short is detected, SensoryGuard records the event, including the detected category, confidence/risk information, timestamp, and intervention taken.
- **Parent report:** At the end of the monitoring session, the recorded events are summarized into a parent report showing the total Shorts analyzed, detected potentially harmful Shorts, detected categories, timestamps, and actions taken.



## 7. Models

The project will evaluate the following models and methods for the different analysis tasks:

- **Video model: VideoMAE**
  VideoMAE will be evaluated as the main video-classification model because it is designed for video understanding and publicly available fine-tuned violence-detection checkpoints can be used as starting points. One available checkpoint performs binary violent/non-violent classification using 16-frame clips at 224×224 resolution. Its reported performance is based on a Bus Violence test set, so it will not be assumed to work directly on YouTube Shorts. It will first be tested on the project's selected data and fine-tuned if required.
- **Speech-to-text: Whisper**
  Whisper will be used to convert spoken parts of a Short into text. Its multilingual models support speech recognition in multiple languages, including Urdu. The resulting transcript will be passed to the language-classification stage. The model size will be selected according to the required accuracy and Android device limitations.
- **Language model: mBERT**
  Multilingual BERT (mBERT) will be evaluated for offensive/toxic-language classification across English, Urdu, and Roman Urdu. XLM-R will be kept as a comparison model if development time and device resources allow. XLM-R was trained across 100 languages and has shown improvements over mBERT on several cross-lingual benchmarks, including improvements reported for Urdu, making it a useful comparison for this project.
- **Visual signals: OpenCV**
  OpenCV will be used to calculate frame-difference and shot-change-related measures. These features will act as supporting visual signals alongside the video-classification model rather than being treated as a separate harmful-content detector.
- **Audio signals: librosa**
  Librosa will be used to extract audio features such as RMS energy and changes in peak level. These measurements will be treated as supporting signals, not as direct evidence that a Short is harmful. Librosa provides RMS and other frame-level audio feature extraction functions.
- **Decision layer**
  The outputs from the video model, visual signals, audio signals, and language classifier will be combined using a simple weighted rule or a small classifier. A threshold and temporal-confirmation rule will then determine whether intervention is required. The threshold will be selected using validation data rather than chosen only by guesswork.
- **Model selection and Android testing**
  Final model selection will consider both classification performance and practical Android performance. Candidate models will be compared using accuracy-related

measures, false-positive rate, inference time, memory use, and end-to-end detection delay on the target device. A model will not be selected only because it has the highest classification score; it must also be practical for the continuous analysis of YouTube Shorts.

## 8. Tools & Technologies

| Area | Tools / Technologies |
|---|---|
| **Application development** | React Native, JavaScript/TypeScript, Visual Studio Code |
| **Android native integration** | Kotlin, React Native Native Modules / Turbo Native Modules, Android SDK |
| **Screen & audio capture** | MediaProjection, Android playback-audio capture, foreground service |
| **Intervention / overlay** | AccessibilityService, accessibility overlay, gesture/swipe interaction |
| **Video processing** | Python, OpenCV, FFmpeg where required |
| **Audio processing** | librosa, NumPy |
| **Speech-to-text** | Whisper / compatible speech-to-text implementation |
| **NLP** | Hugging Face Transformers, mBERT; XLM-R for comparison |
| **Video model** | VideoMAE / Hugging Face Transformers |
| **On-device model deployment** | PyTorch / ExecuTorch for Android |
| **Model training & fine-tuning** | PyTorch, Google Colab or local GPU |
| **Data handling & evaluation** | Pandas, NumPy, scikit-learn |
| **Local storage & reporting** | SQLite / Android Room, React Native report interface |
| **Version control** | GitHub |

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